★ WEEK 10 · PROBLEM SET II

Three Hands-On Problems

Finance Applications · Score a Strategy · Purge a Timeline · Price the Luck

Revised edition — purging both sides, López de Prado's expected maximum, DSR and MinTRL

Each problem: Statement → Principles & Lab Guide → Hand Calculation → Claude Code Prompt.
Small numbers you can compute with a pencil — then a metaprompt that scales the idea to real data.

HOW TO USE

Every Problem Follows the Same Four Steps

Work top to bottom — do not skip the hand calculation; it is where the understanding lives

1 · Statement

the question

A concrete finance scenario with a tiny dataset you can see at a glance.

one slide

2 · Principles

the tools

The formulas involved and a short lab guide: what to compute, in what order.

one slide

3 · Hand Calc

pencil first

Work every number by hand in a table. Small n, clean arithmetic.

1–2 slides

4 · Claude Code

scale it up

A metaprompt: paste it into Claude Code and it builds the full lab on real data.

one slide

Hand first, machine second — you check the machine, never the other way around.

Where the Data Comes From

Public or in-script sources first; CSV files are hand-calc references or offline fallbacks

Problem	Primary source (public · no signup)	Provided fallback file
P1 · Sharpe	numbers on the slides — pure pencil work	fml_w10_p1_returns.csv
P2 · Purged CV	yfinance series (AAPL, 3y, free)	fml_w10_p2_timeline.csv
P3 · Luck, DSR, MinTRL	random signals generated in-script	fml_w10_prices_offline.csv
Real-data extension	any yfinance ticker you like	—

Course rule: free online data (yfinance, FRED, OpenML) or in-script data comes first.

P1

PROBLEM 1 · SHARPE

Score the Strategy First

Five monthly returns — mean, wobble, annualize; then ask whether five months can support the score

A Strategy Reports Five Months of Returns

Compute its Sharpe ratio, monthly and annualized — then face the uncomfortable follow-up

Month	1	2	3	4	5
Return	+2%	−1%	+3%	0%	+1%

Helpers

- Use the population std (divide by n = 5).
- $\sqrt{2} \approx 1.414 \cdot \sqrt{5} \approx 2.236 \cdot \sqrt{12} \approx 3.464$
- Risk-free rate: assume 0 for this toy.

Your deliverables

- Mean return and the five deviations.
- Volatility, monthly Sharpe, annualized Sharpe.
- Follow-up: this was the best of 100 tries — does 2.45 still impress?

The Tool: Reward per Unit of Wobble

One fraction, then a square root to change the clock

$$\mathrm{SR}_{\text{monthly}} = \frac{\bar{r}}{\sigma_r} \qquad \mathrm{SR}_{\text{annual}} = \sqrt{12} \times \mathrm{SR}_{\text{monthly}}$$

Reading it

- Numerator: the average monthly earn.
- Denominator: the typical wobble around that average.
- Annualize by $\sqrt{12}$ because variance adds across months.

Lab guide

- Step 1: mean, then each month's deviation.
- Step 2: σ = $\sqrt{}$(mean of squared deviations).
- Step 3: SR = mean ÷ σ, then × $\sqrt{12}$.

From Five Numbers to One Yardstick

Every step visible — mean, deviations, σ, and the two Sharpes

Step	Arithmetic	Result
Mean	(2 − 1 + 3 + 0 + 1) ÷ 5	1%
Deviations	+1, −2, +2, −1, 0	squares: 1, 4, 4, 1, 0
Volatility σ	$\sqrt{}$(10 ÷ 5) = $\sqrt{}$2	≈ 1.41%
Sharpe (monthly)	1 ÷ 1.41	0.71
Sharpe (annualized)	0.71 × 3.464	≈ 2.45

$$\bar{r} = 1\%, \quad \sigma_r = \sqrt{2}\% \approx 1.41\% \;\Rightarrow\; \mathrm{SR}_{\mathrm{monthly}} = 0.71, \quad \mathrm{SR}_{\mathrm{annual}} \approx 2.45$$

The Follow-Up: Luck's Par for Five Months

Best of N = 100 zero-skill tries · only T = 5 monthly observations · γ ≈ 0.58 (see P3 for the formula)

Step	Arithmetic	Result
Best of 100 draws (noise units)	$0.42 \times 2.33 + 0.58 \times 2.68$	2.53
Noise of one monthly Sharpe	$1 \div \sqrt{5}$	0.447
Monthly par	2.53×0.447	1.13
Annualized par	1.13×3.464	≈ 3.9
Verdict on 2.45	2.45 vs 3.9	BELOW par

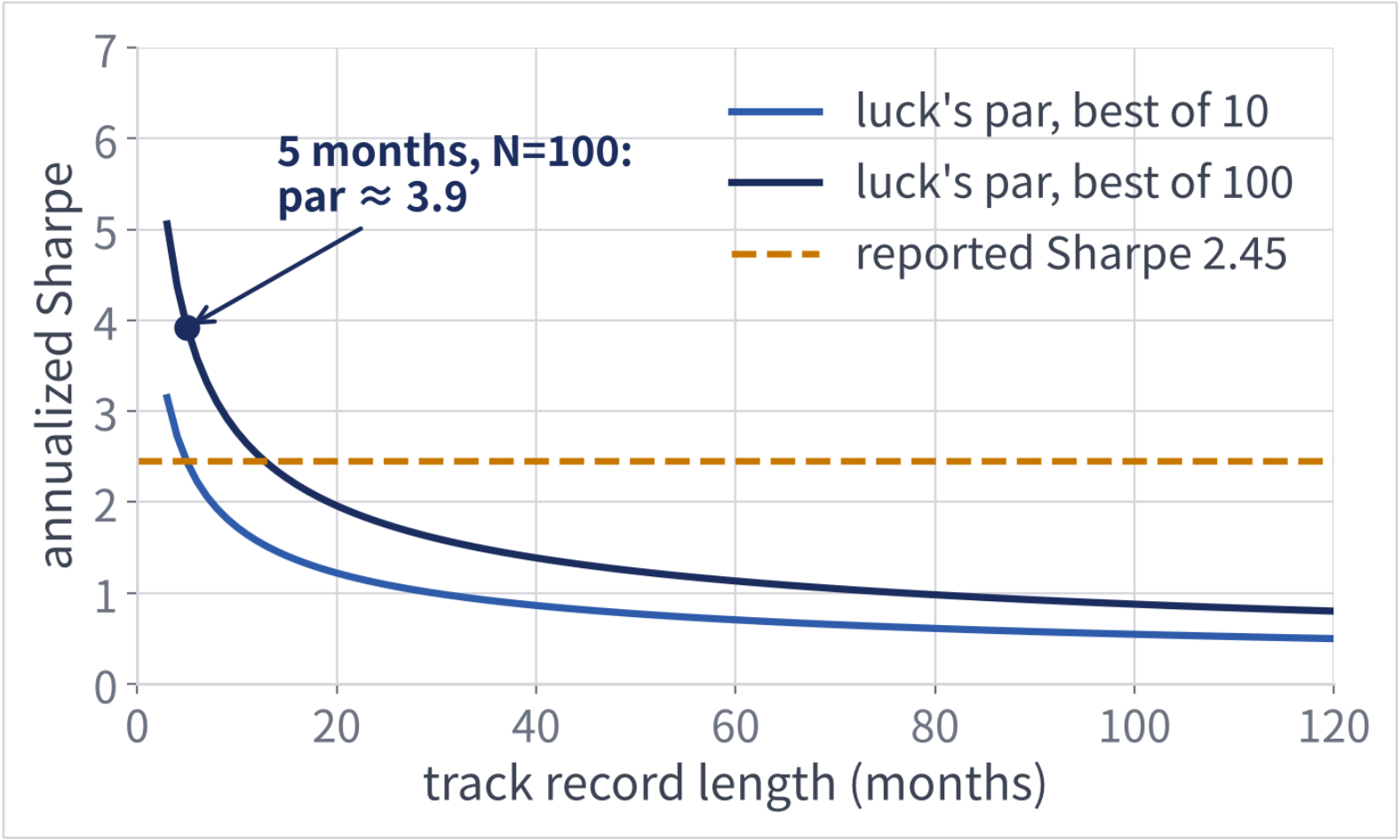
$$\text{par} = \frac{2.53}{\sqrt{5}} \times \sqrt{12} = 1.13 \times 3.46 \approx 3.9 \; > \; 2.45$$

A Short Track Record Raises the Bar for Luck

Luck's par for the best of N strategies falls only as the record lengthens

Read the curve

- Five months: par ≈ 3.9.
- 2.45 clears par only after about 13 months.
- Three years: par ≈ 1.5.



Sharpe 2.45 Is Arithmetic, Not Yet Evidence

The number is correct; whether it means anything depends on N, on T, and on Problems 2 and 3

2.45
annualized Sharpe — the advertisement

3.9
luck's par, best of 100 over 5 months

13 mo
record needed for 2.45 to clear that par

?
survives leakage and deflation — the evidence

Why $\sqrt{12}$ and not 12

- Means grow with time; wobbles grow with $\sqrt{\text{time}}$.
- So the ratio scales by $\sqrt{12}$ monthly, $\sqrt{252}$ daily.

The follow-up answered

- Best of 100 over 5 months: 2.45 falls short of 3.9.
- Few observations make every Sharpe noisy, so luck reaches higher.

Scale It: Sharpe with Honest Error Bars

Paste the metaprompt — Claude Code scores a real series and shows how shaky Sharpe is

[INPUT] Prompt · Claude Code (desktop app, Code tab)

```
Build a single Colab-ready Python script that:
1) downloads AAPL with yfinance (period='3y',
   auto_adjust=True); ret = pct_change().dropna();
   (offline fallback: fml_w10_prices_offline.csv)
2) verifies the toy: returns [2,-1,3,0,1] in %,
   population std -> SR_monthly 0.71, annual 2.45;
3) computes luck's par for N=100 with the
   López de Prado expected-max formula at
   T = 5, 13, 36 months; confirm 3.9 at T=5;
4) computes AAPL's daily Sharpe x sqrt(252);
5) bootstrap: resample daily returns 2000 times,
   plot the Sharpe histogram, 5th-95th band;
6) three-sentence verdict on N, T and the band.
```

[OUTPUT] What to expect

```
toy check: 0.71 / 2.45 exactly

par (N=100): 3.9 / 2.4 / 1.5
  at 5 / 13 / 36 months

AAPL 3y Sharpe: one number
bootstrap band: wide,
  often +/- 0.5 or more

Lesson: the same Sharpe
means less when the record
is short or the trials many.
```

P2

PROBLEM 2 · PURGED CV

Quarantine the Border Days

Ten days, three-day labels — purge on both sides of the test block, then embargo

Ten Days, One Honest Split

Day t's label uses days t to t+2 · test = days 5–7 · embargo = 1 day · classify every day

Day	1	2	3	4	5	6	7	8	9	10
Label uses	1–3	2–4	3–5	4–6	5–7	6–8	7–9	8–10	9–11	10–12

The rules

- TEST: days 5, 6, 7 — their labels reach day 9.
- PURGE any training day whose label overlaps days 5–9, before or after.
- EMBARGO 1 day after the last test label ends.

Your deliverables

- A verdict per day: train, purge, test or embargo.
- Why days 3–4 and 8–9 fall, and why day 10 sits out.
- The honest training count versus naive CV.

The Tool: One Overlap Rule, One Buffer

Compare label windows with the test labels' span, not with the test days alone

$$\text{test information span} = \left[\min_{j \in \text{test}} t_j,\ \max_{j \in \text{test}} (t_j + 2)\right] = [5,\ 9]$$

$$\text{PURGE day } t \text{ if } [t,\ t+2] \cap [5,\ 9] \neq \varnothing; \qquad \text{EMBARGO } h \text{ days after day } 9$$

Why purge both sides

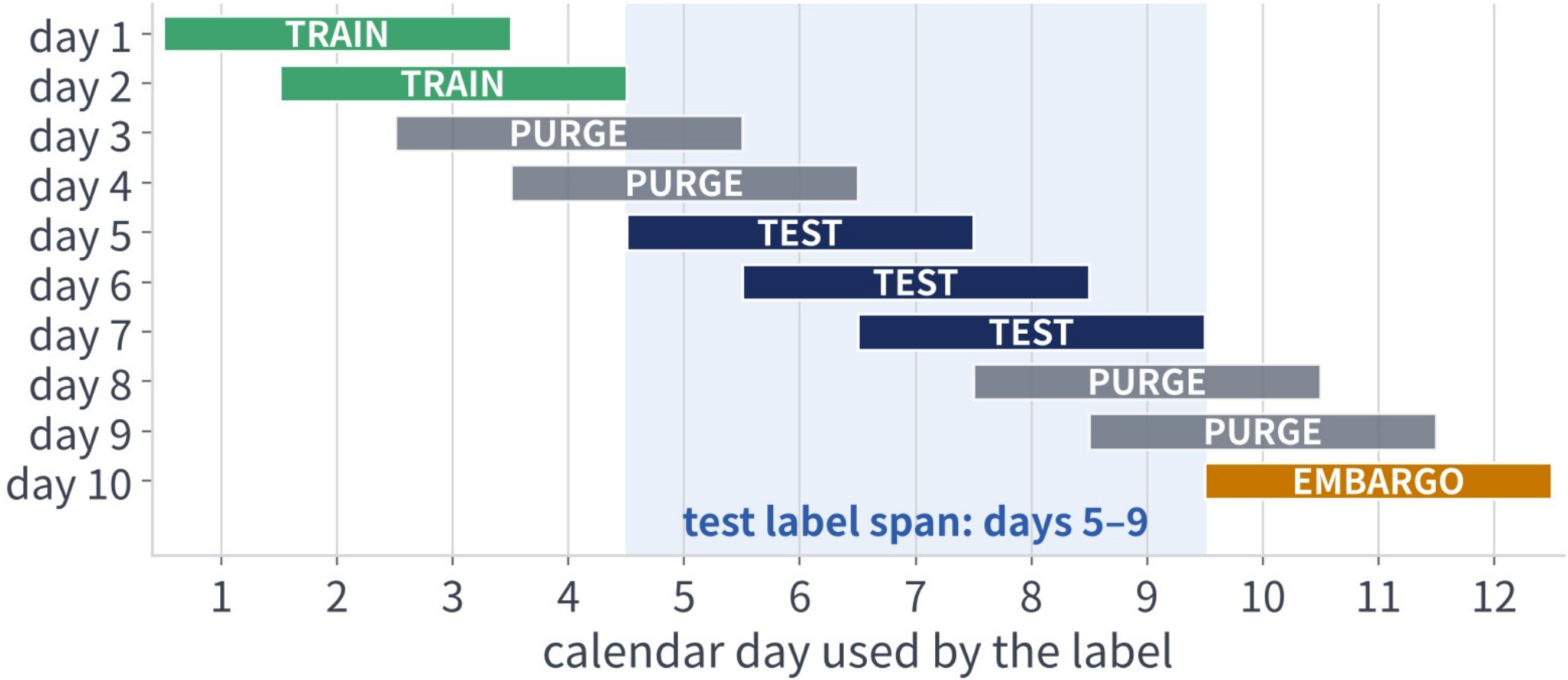
- Day 4's label reaches into the test block.
- Day 8's label shares days 8–9 with test labels.

Why embargo on top

- Purging removes shared labels only.
- Serially correlated features need h more days.

Every Window That Enters the Shaded Span Is Purged

Test labels use days 5–9; days 3, 4, 8, 9 overlap it, and day 10 is the one-day embargo



Ten Verdicts, One Honest Split

Two days survive as training — naive CV would happily use seven

Day	1	2	3	4	5	6	7	8	9	10
Window	1–3	2–4	3–5	4–6	5–7	6–8	7–9	8–10	9–11	10–12
Overlap with 5–9	none	none	5	5–6	—	—	—	8–9	9	none
Verdict	train	train	PURGE	PURGE	test	test	test	PURGE	PURGE	EMBARGO

$$\text{test} = \{5, 6, 7\} \quad \text{purged} = \{3, 4, 8, 9\} \quad \text{embargo} = \{10\} \quad \text{honest train} = \{1, 2\}$$

Day 10 misses the test span, so purging keeps it; the embargo removes it.

Honest Training Set: {1, 2}

Five of the seven non-test days were unusable — and that is the normal case, not the exception

2 / 7

honest training days vs what naive CV uses

4

days purged — two before the test, two after

1

day embargoed after the test labels end

↓ then ✓

scores drop after purging, and become believable

Scaling up

- Purge: any train window touching the test label span, however many days.
- Embargo: h more days after that span, set separately from the label length.

The mindset

- Losing rows feels wasteful; keeping them is fraud.
- An honest 2-day score beats a leaky 7-day score.

Scale It: Build Purged CV and Measure the Leak

Paste the metaprompt — Claude Code implements the quarantine and prices the difference

[INPUT] Prompt · Claude Code (desktop app, Code tab)

```
Build a single Colab-ready Python script that:
1) downloads AAPL with yfinance (period='3y'); builds
   features = two lagged returns, label = sign of
   next 3-day cumulative return; store each row's
   label end date t1 (offline fallback CSV allowed);
2) scores LogisticRegression three ways:
   (a) shuffled KFold(5), (b) TimeSeriesSplit(5),
   (c) purged 5-fold: drop training rows whose
   [t, t1] overlaps the test fold's label span on
   BOTH sides, then embargo 5 days after max t1;
3) verifies on the 10-day toy: train = {1, 2};
4) prints accuracies and rows purged per fold;
5) reruns (c) for horizons [1, 3, 10];
6) three-sentence verdict on trustworthiness.
```

[OUTPUT] What to expect

```
toy check: purged {3,4,8,9},
  embargo {10}, train {1,2}

shuffled: highest score,
  least trustworthy
time-ordered: lower
purged + embargo: lowest,
  honest

purged rows grow with
  the label horizon
```

P3

PROBLEM 3 · LUCK'S PAR

How Good Is Pure Luck?

The best-of-N Sharpe that zero skill earns — computed, deflated, then turned into a data requirement

Three Research Teams, One Question

Best annualized Sharpe on 3 years of daily data (T = 756) — how many tries did luck get?

Team	Strategies tested N	Best annualized Sharpe	Claim
A	10	2.0	'a genuine edge'
B	100	2.0	'a genuine edge'
C	1,000	2.0	'a genuine edge'

Helpers

- $\gamma \approx 0.58$, so $1 - \gamma \approx 0.42$.
- Φ^{-1} pairs: N=10 → 1.28, 1.79 · N=100 → 2.33, 2.68 · N=1,000 → 3.09, 3.38
- $\sqrt{756} \approx 27.5$ · $\sqrt{252} \approx 15.87$

Your deliverables

- Luck's par for each N.
- A verdict per team on Sharpe 2.0.
- The honest reporting rule.

The Tool: the Expected Best of N Flukes

False Strategy Theorem — Bailey & López de Prado (2014), the benchmark inside the DSR

$$\mathbb{E}\left[\max_n \widehat{\mathrm{SR}}_n\right] \approx \sqrt{V[\widehat{\mathrm{SR}}_n]}\left[(1-\gamma)\,\Phi^{-1}\left(1-\frac{1}{N}\right)+\gamma\,\Phi^{-1}\left(1-\frac{1}{Ne}\right)\right]$$

$$\gamma \approx 0.5772\ \text{(Euler–Mascheroni)}, \qquad \sqrt{V[\widehat{\mathrm{SR}}_n]} \approx \frac{1}{\sqrt{T}}\ \text{under zero skill}$$

Reading it

- Zero-skill Sharpe is noise of size $1/\sqrt{T}$.
- The bracket: how far out the best of N sits.
- It climbs with N forever, but slowly.

Lab guide — per team

- Step 1: bracket from the helpers.
- Step 2: ÷ 27.5 → daily par.
- Step 3: × 15.87 → annual par; compare to 2.0.

The Ladder, Then the Judgments

Same Sharpe 2.0 — the margin over luck shrinks from wide to thin

Team	Bracket (noise units)	Daily par (÷ 27.5)	Annual par (× 15.87)	Sharpe 2.0 verdict
A · N=10	0.42 × 1.28 + 0.58 × 1.79 = 1.58	0.057	≈ 0.91	well above par
B · N=100	0.42 × 2.33 + 0.58 × 2.68 = 2.53	0.092	≈ 1.46	above par — dig deeper
C · N=1,000	0.42 × 3.09 + 0.58 × 3.38 = 3.25	0.118	≈ 1.88	barely above — thin

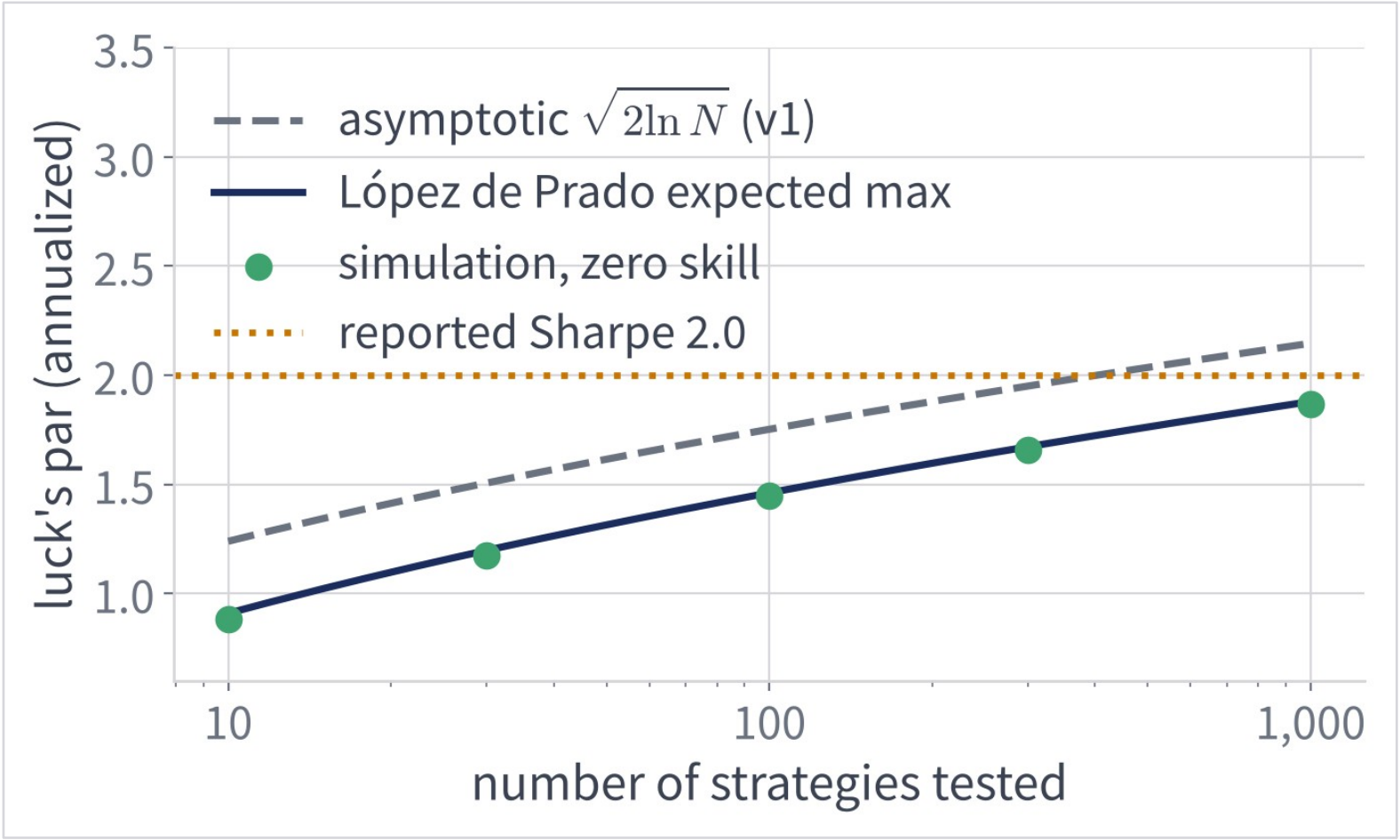
$$N = 10:\ 0.91 \qquad N = 100:\ 1.46 \qquad N = 1000:\ 1.88 \qquad (\text{luck's par},\ T = 756)$$

The Exact Formula Tracks the Simulation

√(2 ln N) is an asymptotic upper bound — at N = 1,000 it still overstates luck by about 15%

What to see

- Green dots: zero-skill simulation.
- Navy line sits on the dots.
- Dashed shortcut flips Team C's verdict.



The Trial Count Is Part of the Result

A Sharpe without its N and its T is an advertisement, not evidence

0.91

luck's par after 10 tries — 2.0 clears it easily

1.88

luck's par after 1,000 tries — 2.0 barely clears it

ln N

the par never stops climbing

SR + N + T

the honest report: all three together

Beating par is not enough

- Par is luck's average best; about half of lucky winners beat it.
- We need a probability: that is the DSR.

The cultural fix

- Log every trial before running it.
- Week 4's factor zoo was this problem at industry scale.

Deflation Turns the Margin into a Probability

Deflated Sharpe Ratio — Bailey & López de Prado (2014) · use daily, non-annualized Sharpe

$$\mathrm{DSR} = \Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{\mathrm{SR}}^2}}\right)$$

$$\mathrm{SR}^* = \text{luck's par (daily)}, \qquad \hat{\gamma}_3 = \text{skewness}, \quad \hat{\gamma}_4 = \text{kurtosis (normal: } 0,\ 3)$$

Reading it

- Top: margin over luck, scaled by $\sqrt{T}$.
- Bottom: negative skew and fat tails widen the noise.

What it fixes

- More trials raise SR*.
- Shorter records shrink $\sqrt{(T-1)}$.

Deflating Sharpe 2.0 for the Three Teams

Normal returns: denominator ≈ 1.004, treat it as 1 · par from unrounded brackets, four decimals

Step	Team A	Team B	Team C
Daily Sharpe: 2.0 ÷ 15.87	0.1260	0.1260	0.1260
Daily luck's par SR*	0.0573	0.0920	0.1184
Margin over luck	0.0687	0.0340	0.0076
Margin × $\sqrt{}$755 (≈ 27.48)	1.89	0.93	0.21
DSR = Φ(·)	0.97	0.82	0.58

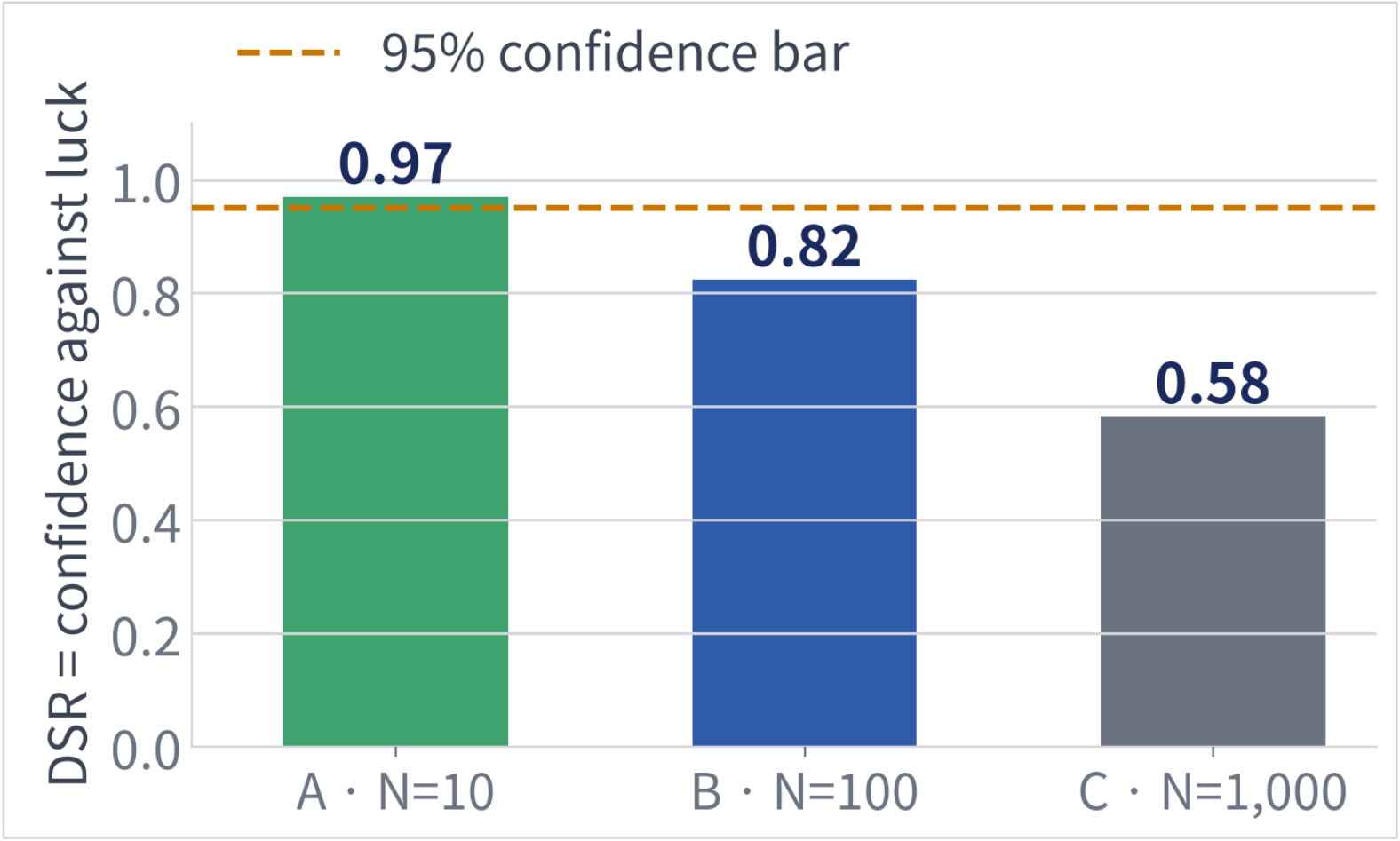
$$\mathrm{DSR}_A \approx 0.97 \qquad \mathrm{DSR}_B \approx 0.82 \qquad \mathrm{DSR}_C \approx 0.58$$

Only Team A Clears a 95% Confidence Bar

Same Sharpe 2.0, same three years — confidence against luck falls from 0.97 to 0.58 as N grows

Verdicts

- A: beats luck at 95%.
- B: promising, not proven.
- C: no better than luck.
- Pure luck alone scores near 0.5.



Minimum Track Record Length Says How Long to Wait

Bailey & López de Prado (2012) · invert the DSR: how many days until 95% confidence (z = 1.645)?

$$\text{MinTRL} = 1 + \left(1 - \hat{\gamma}_3 \widehat{\text{SR}} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{\text{SR}}^2\right) \left(\frac{z_\alpha}{\widehat{\text{SR}} - \text{SR}^*}\right)^2$$

Team	Margin	Days: (1.645 ÷ margin)²	Years (÷ 252)	Have 3 years?
A · N=10	0.0687	≈ 573	≈ 2.3	yes
B · N=100	0.0340	≈ 2,341	≈ 9.3	no
C · N=1,000	0.0076	≈ 46,850	≈ 186	never in practice

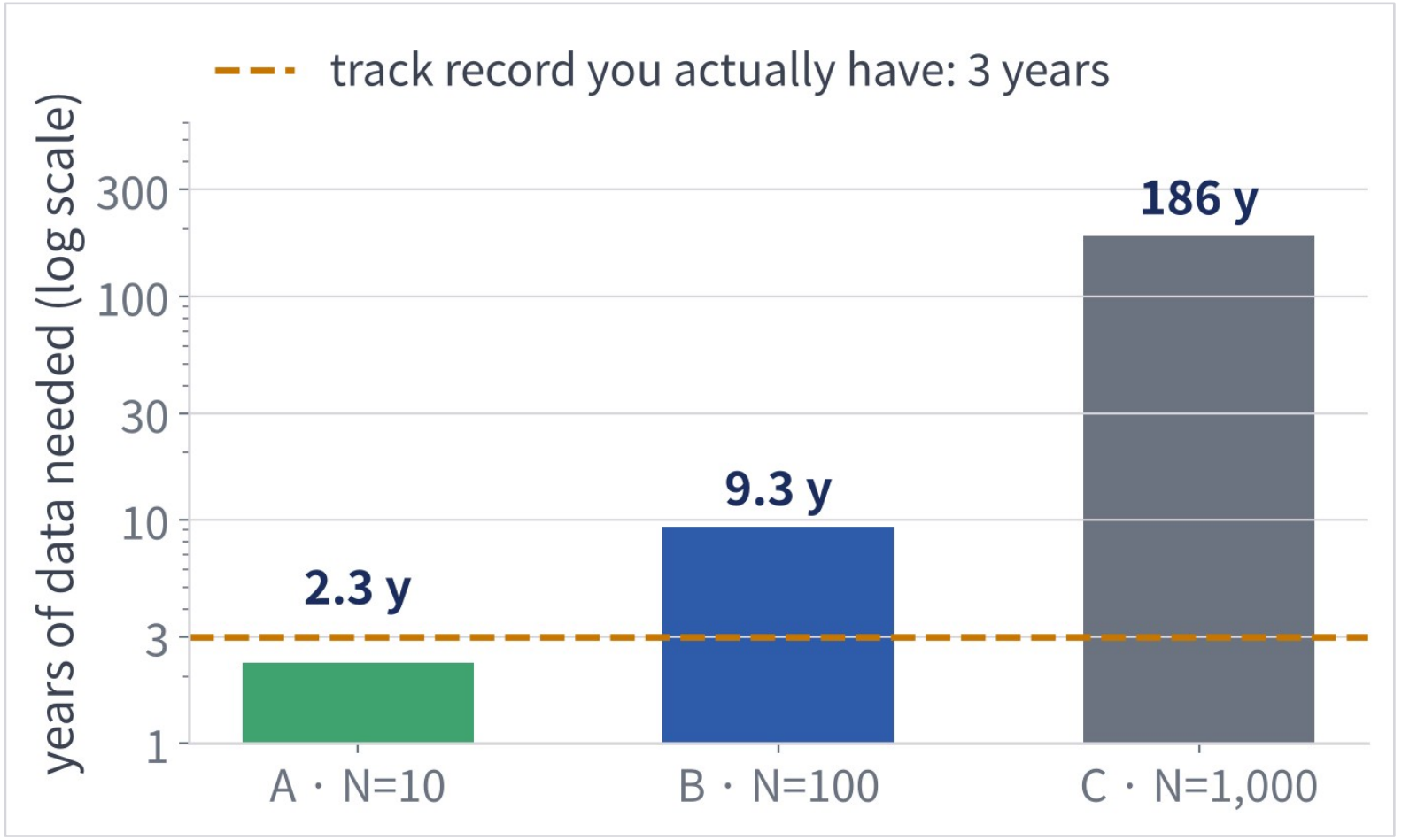
Normal returns; the leading 1 is negligible. Keep four decimals — this formula amplifies rounding.

A Thin Margin over Luck Demands Decades of Data

Years of daily data needed before Sharpe 2.0 is significant at 95%, given each team's N

The lesson

- Short record? Ask for more time.
- Many trials? Ask for even more.
- C cannot be rescued by waiting.



Four More Guards Against Rising N and Shrinking T

DSR and MinTRL price the problem; these practices keep N honest and T untouched

Effective N

cluster the trials

Correlated variants are not independent tries. Cluster trial returns (ONC) and count clusters.

López de Prado (2019)

Multiple testing

raise the hurdle

Holm or BHY corrections; with hundreds of factors, demand t > 3.

Harvey, Liu & Zhu (2016)

PBO · CSCV

measure overfitting

How often does the in-sample winner fall below the median out of sample?

Bailey et al. (2017)

Pre-register

freeze T

Log every trial, hold out untouched data, and test the winner once.

N is a fact, not a memory

Scale It: Simulate the Ladder, Then Deflate

Paste the metaprompt — Claude Code races luck, then issues DSR and MinTRL verdicts

[INPUT] Prompt · Claude Code (desktop app, Code tab)

```
Build a single Colab-ready Python script that:
1) downloads AAPL with yfinance (period='3y');
   ret = pct_change().dropna() (offline CSV ok);
2) for N in [10, 100, 1000]: N random +/-1 signals,
   annualized Sharpe of signal*ret, keep the best;
   repeat 200 times and average;
3) plot simulation vs López de Prado expected max
   and vs sqrt(2 ln N), all x sqrt(252/T);
4) dsr(sr_daily, N, T, skew, kurt) and
   min_trl(...); run on (2.0/sqrt(252), N) for
   N = 10, 100, 1000; confirm 0.97 / 0.82 / 0.58;
5) three-sentence verdict: what to demand from
   anyone reporting a backtest Sharpe.
```

[OUTPUT] What to expect

```
simulation sits on the
  exact formula; sqrt(2 ln N)
  runs about 15-35% high

par: 0.91 / 1.46 / 1.88
DSR: 0.97 / 0.82 / 0.58
MinTRL: 2.3 / 9.3 / 186 y

Demand: Sharpe, N, T,
  skew, kurtosis, DSR.
```

Three Things Your Hands Now Know

You computed every number yourself; the code only repeats what you already understand

1

Sharpe by hand: 0.71 monthly, 2.45 annualized

Correct arithmetic — but best of 100 over five months needs about 3.9 just to beat luck.

2

Purged CV: honest train = {1, 2}

Purge days 3–4 and 8–9 for label overlap on both sides; embargo day 10 after the test labels end.

3

Luck's par 0.91 → 1.46 → 1.88; DSR 0.97 → 0.82 → 0.58

More trials and shorter records both raise the bar; DSR and MinTRL put a price on each.

Hand calculation first — then let Claude Code scale the same idea to real data.

What Changed from the First Edition

For instructors comparing editions — the first edition is kept unchanged

Where	First edition	This edition
P1 follow-up	Par 1.75 (from P3, T = 756): 2.45 clears it	Par ≈ 3.9 for T = 5 months: 2.45 falls short
P2 purge rule	Compared windows with test days 5–7	Compare with test label span 5–9
P2 verdicts	Day 8 embargo; days 9, 10 train	Days 8, 9 purge; day 10 embargo
P2 counts	Train {1, 2, 9, 10}; naive uses 8	Train {1, 2}; naive uses 7
P3 par	$\sqrt{(2 \ln N)}$: 1.24 / 1.75 / 2.15	Expected max: 0.91 / 1.46 / 1.88
P3 Team C	Below par — luck explains it	Barely above; DSR 0.58
Added	—	DSR, MinTRL, four guards